

FSUC4320

## Chloramine-T trihydrate, Sodium salt

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**产品说明:**  
**Product Description:** 三水合氯铵-T  
Chloramine-T trihydrate, Sodium salt

**Cat No. :** C/4320/50, C/4320/53  
**Synonyms** N-Chloro-p-toluenesulfonamide, sodium salt; Tosylchloramide sodium  
**CAS No** 7080-50-4  
**Molecular Formula** C7 H7 Cl N Na O2 S . 3 H2 O

**Supplier** **UK entity/business name**  
Fisher Scientific UK  
Bishop Meadow Road, Loughborough,  
Leicestershire LE11 5RG, United Kingdom

**EU entity/business name**  
Thermo Fisher Scientific  
Janssen Pharmaceuticaaan 3a  
2440 Geel, Belgium

**Emergency Telephone Number** Tel: 01509 231166  
Chemtrec US: (800) 424-9300  
Chemtrec EU: 001-703-527-3887

**E-mail address** begel.sdsdesk@thermofisher.com

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

### SECTION 2. HAZARD IDENTIFICATION

**Physical State**  
Powder Solid

**Appearance**  
Off-white

**Odor**  
Slight chlorine

#### Emergency Overview

Harmful if swallowed. Causes severe skin burns and eye damage. May cause allergy or asthma symptoms or breathing difficulties if inhaled. Contact with acids liberates toxic gas.

#### Classification of the substance or mixture

|                                   |              |
|-----------------------------------|--------------|
| Acute Oral Toxicity               | Category 4   |
| Skin Corrosion/Irritation         | Category 1 B |
| Serious Eye Damage/Eye Irritation | Category 1   |
| Respiratory Sensitization         | Category 1   |

#### Label Elements

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**Signal Word****Danger****Hazard Statements**

H302 - Harmful if swallowed

H314 - Causes severe skin burns and eye damage

H334 - May cause allergy or asthma symptoms or breathing difficulties if inhaled

**Precautionary Statements****Prevention**

P264 - Wash face, hands and any exposed skin thoroughly after handling

P270 - Do not eat, drink or smoke when using this product

P271 - Use only outdoors or in a well-ventilated area

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P284 - In case of inadequate ventilation wear respiratory protection

**Response**

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P310 - Immediately call a POISON CENTER or doctor

P330 - Rinse mouth

P331 - Do NOT induce vomiting

P362 + P364 - Take off contaminated clothing and wash it before reuse

**Storage**

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**

Contact with acids liberates toxic gas.

**Health Hazards**

Harmful if swallowed. Corrosive. Causes skin and eye burns. May cause allergy or asthma symptoms or breathing difficulties if inhaled.

**Environmental hazards**

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants. Will likely be mobile in the environment due to its water solubility. The product is water soluble, and may spread in water systems.

Toxic to terrestrial vertebrates. This product does not contain any known or suspected endocrine disruptors.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| Component               | CAS No    | Weight % |
|-------------------------|-----------|----------|
| Chloramine-T trihydrate | 7080-50-4 | >95      |
| Chloramine-T            | 127-65-1  | -        |

**SECTION 4. FIRST AID MEASURES****General Advice**

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.

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Keep eye wide open while rinsing.

### Skin Contact

Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Call a physician immediately.

### Inhalation

Remove to fresh air. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Call a physician or poison control center immediately. If not breathing, give artificial respiration.

### Ingestion

Do NOT induce vomiting. Immediate medical attention is required. Drink plenty of water. Never give anything by mouth to an unconscious person.

### Most important symptoms and effects

Causes burns by all exposure routes. May cause allergy or asthma symptoms or breathing difficulties if inhaled. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

### Self-Protection of the First Aider

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

### Notes to Physician

Treat symptomatically.

## SECTION 5. FIRE-FIGHTING MEASURES

### Suitable Extinguishing Media

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

### Extinguishing media which must not be used for safety reasons

No information available.

### Specific Hazards Arising from the Chemical

The product causes burns of eyes, skin and mucous membranes.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 6. ACCIDENTAL RELEASE MEASURES

### Personal Precautions

Evacuate personnel to safe areas. Use personal protective equipment as required. Avoid contact with skin, eyes or clothing.

### Environmental Precautions

Do not allow material to contaminate ground water system. Should not be released into the environment. See Section 12 for additional Ecological Information.

### Methods for Containment and Clean Up

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

Refer to protective measures listed in Sections 8 and 13.

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### SECTION 7. HANDLING AND STORAGE

#### Handling

Use only under a chemical fume hood. Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Do not breathe dust. Do not ingest. If swallowed then seek immediate medical assistance.

#### Storage

Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area.

#### Specific Use(s)

Use in laboratories

### SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

#### Control Parameters

#### Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust

#### Exposure Controls

#### Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

#### Personal protective equipment

**Eye Protection** Goggles (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Natural rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Nitrile rubber | recommendations   |                 |             |                       |
| Neoprene       |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Long sleeved clothing

**Respiratory Protection** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Particulates filter conforming to EN 143

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|                                        |                                                                                                                                                                                                                                                                                          |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Small scale/Laboratory use</b>      | Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.<br><b>Recommended half mask:-</b> Particle filtering: EN149:2001<br>When RPE is used a face piece Fit Test should be conducted |
| <b>Hygiene Measures</b>                | Handle in accordance with good industrial hygiene and safety practice.                                                                                                                                                                                                                   |
| <b>Environmental exposure controls</b> | No information available.                                                                                                                                                                                                                                                                |

### SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                                |                               |                                          |
|------------------------------------------------|-------------------------------|------------------------------------------|
| <b>Appearance</b>                              | Off-white                     |                                          |
| <b>Physical State</b>                          | Powder Solid                  |                                          |
| <b>Odor</b>                                    | Slight chlorine               |                                          |
| <b>Odor Threshold</b>                          | No data available             |                                          |
| <b>pH</b>                                      | 8-10                          | 5% aq.sol                                |
| <b>Melting Point/Range</b>                     | 170 - 177 °C / 338 - 350.6 °F |                                          |
| <b>Softening Point</b>                         | No data available             |                                          |
| <b>Boiling Point/Range</b>                     | No data available             |                                          |
| <b>Flash Point</b>                             | 192 °C / 377.6 °F             | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>                        | Not applicable                | Solid                                    |
| <b>Flammability (solid,gas)</b>                | No information available      |                                          |
| <b>Explosion Limits</b>                        | No data available             |                                          |
| <b>Vapor Pressure</b>                          | No information available      |                                          |
| <b>Vapor Density</b>                           | Not applicable                | Solid                                    |
| <b>Specific Gravity / Density</b>              | No data available             |                                          |
| <b>Bulk Density</b>                            | No data available             |                                          |
| <b>Water Solubility</b>                        | 150 g/l (25°C)                |                                          |
| <b>Solubility in other solvents</b>            | No information available      |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                               |                                          |
| <b>Component</b>                               | <b>log Pow</b>                |                                          |
| Chloramine-T                                   | -1.3                          |                                          |
| <b>Autoignition Temperature</b>                | Not applicable                |                                          |
| <b>Decomposition Temperature</b>               | No data available             |                                          |
| <b>Viscosity</b>                               | Not applicable                | Solid                                    |
| <b>Explosive Properties</b>                    | No information available      |                                          |
| <b>Oxidizing Properties</b>                    | No information available      |                                          |
| <b>Molecular Formula</b>                       | C7 H7 Cl N Na O2 S . 3 H2 O   |                                          |
| <b>Molecular Weight</b>                        | 281.69                        |                                          |

### SECTION 10. STABILITY AND REACTIVITY

|                                 |                                          |
|---------------------------------|------------------------------------------|
| <b>Stability</b>                | Stable under normal conditions.          |
| <b>Hazardous Reactions</b>      | None under normal processing.            |
| <b>Hazardous Polymerization</b> | Hazardous polymerization does not occur. |
| <b>Conditions to Avoid</b>      | Incompatible products. Excess heat.      |
| <b>Materials to avoid</b>       | Acids. Strong oxidizing agents.          |

**Hazardous Decomposition Products** Nitrogen oxides (NOx). Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Chlorine. Sulfur oxides. Hydrogen chloride gas.

### SECTION 11. TOXICOLOGICAL INFORMATION

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### Product Information

#### (a) acute toxicity;

| Component    | LD50 Oral                | LD50 Dermal                  | LC50 Inhalation                          |
|--------------|--------------------------|------------------------------|------------------------------------------|
| Chloramine-T | LD50 = 935 mg/kg ( Rat ) | LD50 > 2020 mg/kg ( Rabbit ) | LC50 > 275 mg/m <sup>3</sup> ( Rat ) 4 h |

(b) skin corrosion/irritation; Category 1 B

(c) serious eye damage/irritation; No data available

#### (d) respiratory or skin sensitization;

Respiratory  
Skin

Category 1

No data available

No information available

(e) germ cell mutagenicity;

No data available

Not mutagenic in AMES Test

(f) carcinogenicity;

No data available

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity;

No data available

(h) STOT-single exposure;

No data available

(i) STOT-repeated exposure;

No data available

Target Organs

No information available.

(j) aspiration hazard;

Not applicable  
Solid

#### Symptoms / effects, both acute and delayed

Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

## SECTION 12. ECOLOGICAL INFORMATION

#### Ecotoxicity effects

Do not empty into drains.

| Component    | Freshwater Fish                                                                                                                                                                      | Water Flea                            | Freshwater Algae | Microtox |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|------------------|----------|
| Chloramine-T | LC50: = 31 mg/L, 96h semi-static (Poecilia reticulata)<br>LC50: 20.2 - 26.2 mg/L, 96h flow-through (Oncorhynchus mykiss)<br>LC50: 1.63 - 2.19 mg/L, 96h static (Oncorhynchus mykiss) | EC50: = 4.5 mg/L, 48h (Daphnia magna) |                  |          |

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|  |                                                                |  |  |  |
|--|----------------------------------------------------------------|--|--|--|
|  | LC50: 6.52 - 7.51 mg/L,<br>96h static (Pimephales<br>promelas) |  |  |  |
|--|----------------------------------------------------------------|--|--|--|

**Persistence and Degradability**  
**Persistence** Expected to be biodegradable  
Soluble in water, Persistence is unlikely, based on information available.

**Bioaccumulative Potential** Bioaccumulation is unlikely

| Component    | log Pow | Bioconcentration factor (BCF) |
|--------------|---------|-------------------------------|
| Chloramine-T | -1.3    | No data available             |

**Mobility in soil** The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility Highly mobile in soils

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors  
**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

### SECTION 13. DISPOSAL CONSIDERATIONS

**Waste from Residues/Unused Products** Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point.

**Other Information** Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not flush to sewer. Large amounts will affect pH and harm aquatic organisms.

### SECTION 14. TRANSPORT INFORMATION

#### Road and Rail Transport

**UN-No** UN3263  
**Proper Shipping Name** Corrosive solid, basic, organic, n.o.s.  
**Technical Shipping Name** Chloramine-T, sodium salt  
**Hazard Class** 8  
**Packing Group** III

#### IMDG/IMO

**UN-No** UN3263  
**Proper Shipping Name** Corrosive solid, basic, organic, n.o.s.  
**Technical Shipping Name** Chloramine-T, sodium salt  
**Hazard Class** 8  
**Packing Group** III

#### IATA

**UN-No** UN3263  
**Proper Shipping Name** Corrosive solid, basic, organic, n.o.s.  
**Technical Shipping Name** Chloramine-T, sodium salt  
**Hazard Class** 8  
**Packing Group** III

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**Special Precautions for User** No special precautions required

### SECTION 15. REGULATORY INFORMATION

#### International Inventories

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component               | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL        |
|-------------------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|-------------|
| Chloramine-T trihydrate | -                                                   | -                                       | X    | X     | -         | -    | -   | X     | -    |      | -    | -           |
| Chloramine-T            | -                                                   | -                                       | X    | X     | 204-854-7 | X    | X   | X     | X    | X    | X    | 2000-3-1539 |

#### National Regulations

### SECTION 16. OTHER INFORMATION

**Creation Date** 24-Nov-2010  
**Revision Date** 04-Apr-2024  
**Revision Summary** Not applicable.

#### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

#### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**PNEC** - Predicted No Effect Concentration

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)



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**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**